

Plant Life Cycles

About 2 weeks. 8 lessons. Anchor reading: "From Seed to Seed: The Life of a Bean Plant."

Big idea

Plants grow, change, and make new plants. A bean grows from a seed, makes flowers and new seeds, and the cycle starts again.

Learning goals

1. Name and order the stages of a bean's life cycle.
2. Explain what a seed needs to start growing.
3. Tell how pollination leads to fruit and seeds.
4. Use observations and text to explain the cycle.

Lesson by lesson

Lesson	Focus	Hands-on activity
1	What is a life cycle?	Read the anchor passage. Sort bean plant picture cards into order.
2	Inside a seed	Open soaked lima beans with a hand lens. Find the baby plant.
3	What does a seed need to grow?	The four-cup test: water, light, and soil. Predict and start a log.
4	Germination and seedlings	Check the cups. Measure and sketch roots and stems.
5	Plant parts and their jobs	Stand celery in colored water. Watch the stem carry water up.
6	Flowers and pollination	Be the bee: move chalk "pollen" between paper flowers.
7	Fruits and new seeds	Open a bean pod, a pepper, and an apple. Compare the seeds.
8	The cycle repeats	Final task: draw and label a bean plant life cycle.

Assessment plan

- **Every day:** observation log and an exit question.
- **Mid-unit check (Lesson 4):** five questions on seeds, plus a labeled sketch.
- **Final task (Lesson 8):** draw and label a bean plant's life cycle, with arrows that loop back. Use five vocabulary words. Add one sentence on how the cycle starts again.
- **Checklist:** stages in order, labels correct, arrows make a cycle, sentence explains the repeat.

Materials

Dry lima and pinto beans, clear cups (4 per group), paper towels, potting soil, a spray bottle, hand lenses, rulers, celery, food coloring, pipe cleaners, colored chalk, paper flowers, a bean pod, a pepper, an apple, and observation logs.

Vocabulary

seed: holds a baby plant and its food

germinate: to start to grow from a seed

seedling: a young plant just out of the soil

roots: hold the plant in place and take in water

stem: holds the plant up and carries water

flower: the part of a plant that makes seeds

pollen: a yellow dust made by flowers

pollination: moving pollen so seeds can form

fruit: the part that holds and protects seeds

life cycle: the stages of life, again and again

Family connection

Seed hunt: at a meal, find seeds inside fruits and vegetables. Draw one to share in class. Or try the bean-in-a-cup activity.